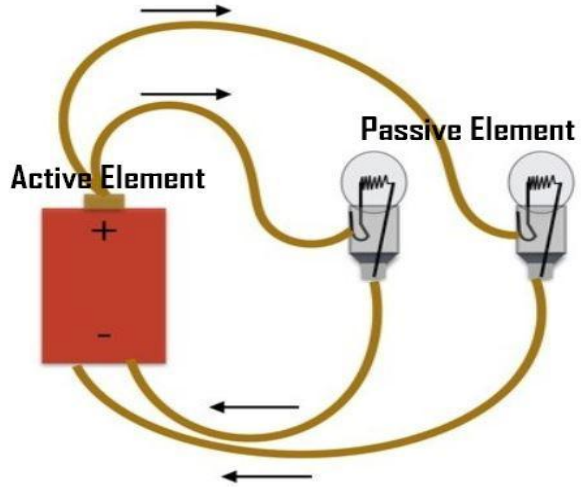
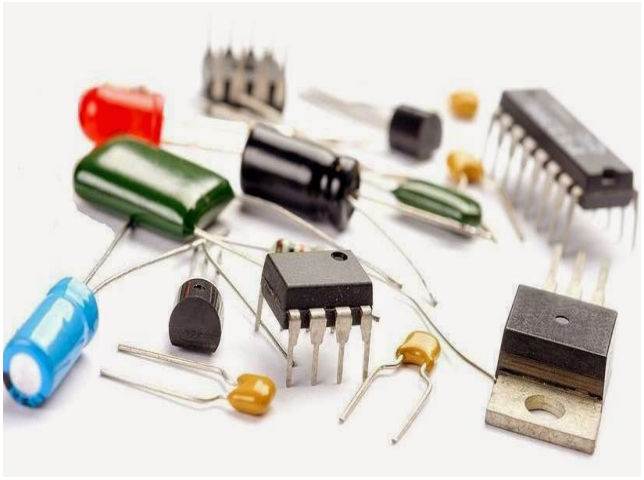
BASIC ELECTRONICS COMPONENTS

Components are classified as Active and Passive components.





Passive Components:

- Those devices or components which do not require an external source to operate in a circuit are called Passive Components. A passive component is an electronic component which can only receive energy, which it can either dissipate, absorb or store it in an

electric field or a magnetic field. Passive elements do not need any form of electrical power to operate.

As the name ‘passive’ suggests – passive devices do not provide gain or amplification. Passive components cannot amplify, oscillate, or generate an electrical signal. They are energy acceptor. They lay in Non-Linear category.

Examples: Resistors, Capacitors, Inductors, Transformer, Motors etc.

Active Components :

Active device transforms and injects power or energy into a circuit. Active element produces energy in the form of voltage or current. They provide power gain (Amplifier). They are energy donor. They lay in linear category.

Examples: Diode, Transistor, SCR, ICs, DC generator, Current & Voltage sources etc.

Let us see all components in detail

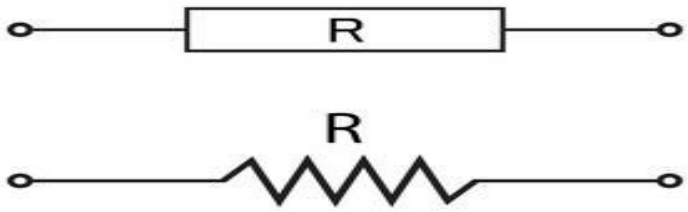
Resistors:

A passive electrical component with two terminals that are used for either limiting or regulating the flow of electric current in electrical circuits. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements**.** Resistor works based on the principle of Ohm’s law which states that “voltage applied across the terminals of a resistor is directly proportional to the current flowing through it”

$$V=IR$$

SI-unit : Ohm Ω, Kilo ohms $K\Omega$, megaohms ($M\Omega$), milliohm and so on are known as the higher multiple and sub-multiple values of Ohm.

Symbol:



Power Dissipation:

Any resistor in a circuit that has a voltage drop across it dissipates electrical power. This electrical power is converted into heat energy hence all resistors have a power rating. This is the maximum power that can be dissipated from the resistor without it burning out. The rate of conversion is the power of dissipation.

- If P is the power in watts dissipated by the resistor, in a DC circuit:

$$P = V * I$$

- By substitution in Ohm's law, we can express power (watts) in terms of current and resistance:

$$P = I^2 * R$$

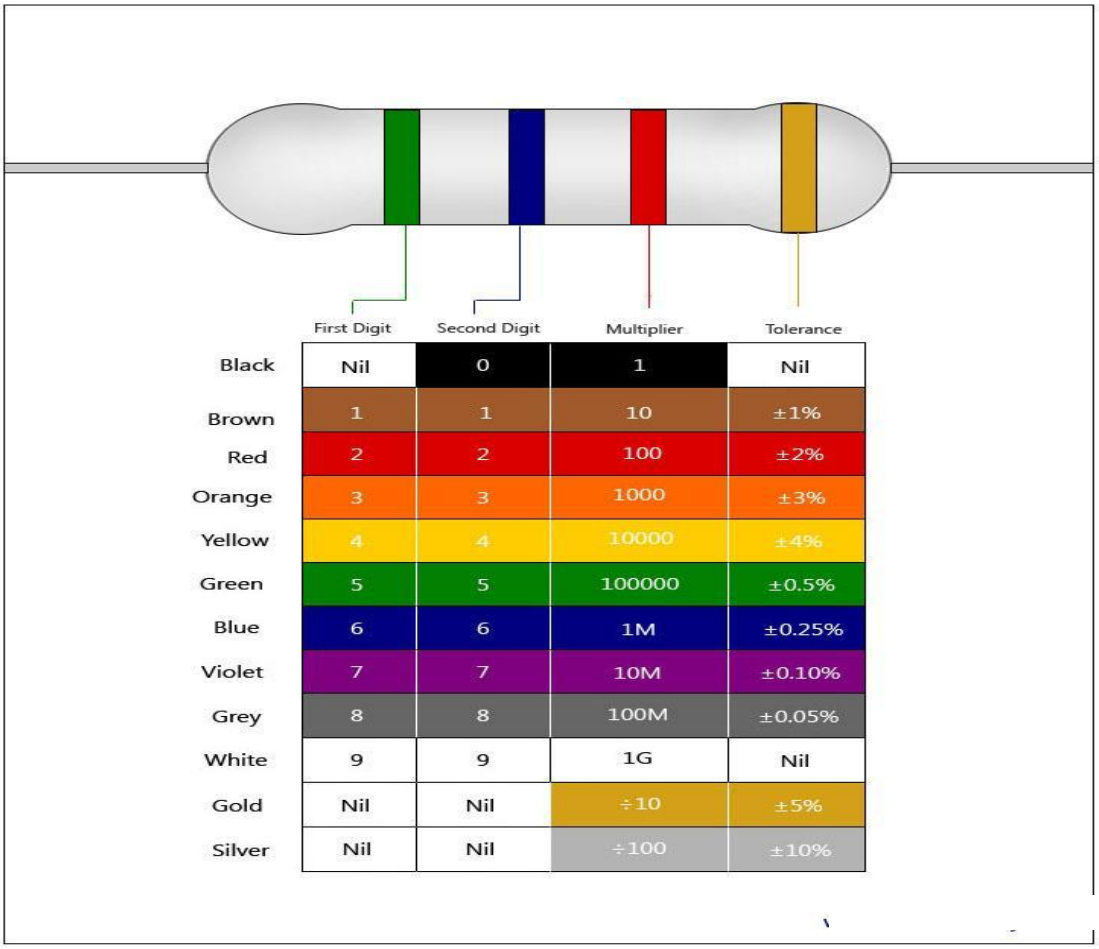
- We can also express power (watts) in terms of voltage and resistance:

$$P = V^2 / R$$

Colour Code Chart:

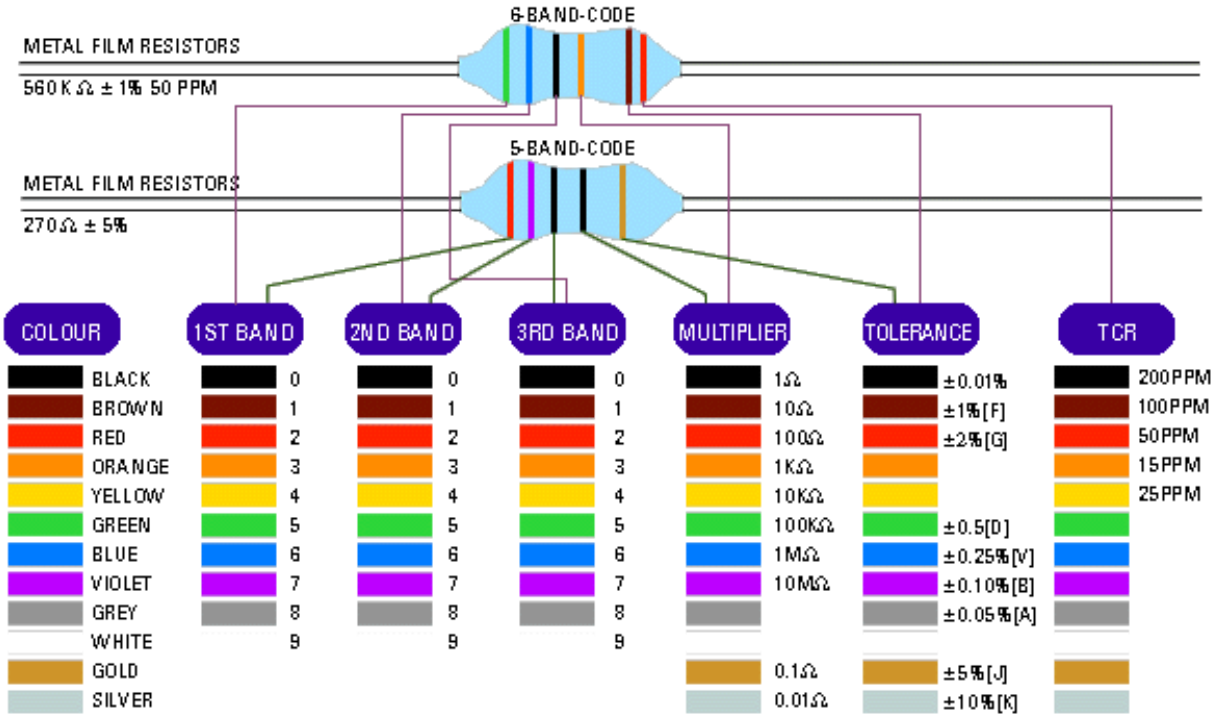
For 4 band Resistor:

4 Band Resistor

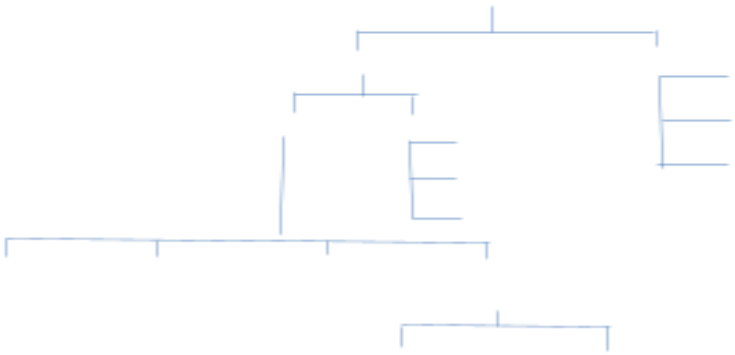


	First Digit	Second Digit	Multiplier	Tolerance
Black	Nil	0	1	Nil
Brown	1	1	10	±1%
Red	2	2	100	±2%
Orange	3	3	1000	±3%
Yellow	4	4	10000	±4%
Green	5	5	100000	±0.5%
Blue	6	6	1M	±0.25%
Violet	7	7	10M	±0.10%
Grey	8	8	100M	±0.05%
White	9	9	1G	Nil
Gold	Nil	Nil	÷10	±5%
Silver	Nil	Nil	÷100	±10%

For 5 and 6 band Resistor



Types of Resistors:



- **Linear Resistor:** Resistor whose value change with change in applied voltage and temperature.
- **Fixed Resistor**: Resistors with specific values and values cannot be changed .
- **Carbon Composition Resistor**: Cheap ,General Purpose Resistors. Their resistive element is manufactured from a mixture of finely ground carbon dust or graphite (similar to pencil lead) and a non-conducting ceramic (clay) powder to bind it all together. This is the most common type of low wattage resistor. These resistors are cheap and reliable and stability is high.

Wire wound resistor: The conductive wire can be made of varying alloys and thickness to control the resistance value. Typically used in high power and industrial applications such as circuit breakers and fuses. Wirewound resistors have some characteristics that are attractive in specialist applications, such as high precision instrumentation where tolerances much better than 0.01% and very low-temperature coefficients (TCR) are required. However, it's worth noting that as they consist of wire wound around an insulated core, which is an inductor by definition, these resistors are not recommended for high-frequency applications, and are generally not available in surface mount form.



Thick film Resistors: Produced by applying a resistive film or paste, a mixture of glass and conductive materials, to a substrate. Applications : Test and measurement equipment, monitoring equipment, medical equipment, audio applications, precision controls and instrumentation.



- **Thin film Resistors:**

- **Carbon Film Resistors:** Constructed by forming a carbon film onto a ceramic substrate. Advantages are the film is largely non-inductive, makes them useful for high-frequency applications . Ceramic is an excellent insulator for heat and electricity. The layout has a large cross-section which lends itself to higher operating voltages and better survivability to transients than many other types.



- **Metal film Resistor**: made by placing a layer of metals such as ruthenium onto an insulating substrate. Available in a large number of types and packages and compared to carbon film resistors have largely replaced carbon film resistors in most standard applications due to their lower noise, tighter tolerances and generally better temperature coefficients.



Variable type Resistor:

- **Potentiometer:** Potentiometer is a three terminal device which is used for controlling the level of voltage in the circuit. The resistance between two external terminals is constant while the third terminal is connected with moving contact (Wiper) which is variable. The value of resistance can be changed by rotating the wiper which is connected to the control shaft. It functions as a potential divider, which is used to generate a voltage signal depending on the location of the potentiometer.

- Applications: amplifier gain control, measurement of the distance or angles, tuning of the circuits and many more.

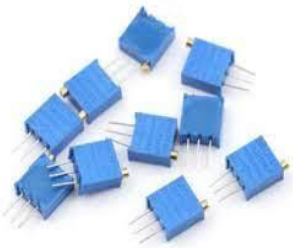


- R**heostat:** Rheostats are a two or three terminal device which is used for the current limiting purpose by hand or manual operation. Rheostats are also known as tapped resistors or variable wire wound resistors. These are adjustable resistor used in applications that require the adjustment of current or the varying of resistance in an electric circuit. The rheostat can adjust generator characteristics, dim lights, and start or control the speed of motors.



- **Presets:** A preset is a three legged electronic component. Made to offer varying resistance in a circuit. The resistance is varied by adjusting the rotary control over it. The adjustment can be done by using a small screw driver or a similar tool. The resistance does not vary linearly but rather varies in exponential or logarithmic manner. Such variable resistors are commonly used for adjusting sensitivity along with a sensor.







Non -Linear Resistors:

Thermistor: Thermal means temperature. In this resistor, the resistance varies with temperature. If heat increases, the resistance decreases and vice versa. This is used for measurement and control purposes.

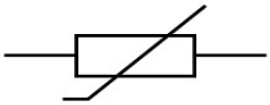
The main types of thermistors are **NTC** and **PTC**.

- NTC is **Negative Temperature Coefficient** and in such devices, the resistance decreases as the temperature increases. These are used to protect the devices from over-voltage conditions.
- PTC is **Positive Temperature Coefficient** and in such devices, the resistance increases as the temperature increases. These are used to protect the devices from over current conditions.

The following figure shows an NTC thermistor, along with its symbol.



Image of a Thermistor (NTC)

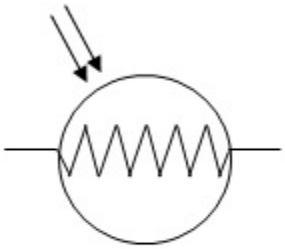


Symbol for Thermistor

Light Dependant resistor(photo Resistor): Photo means light. In this resistor, the resistance varies with light. As light increases resistance decreases and vice versa. This is also used for measurement and control purposes. It is also called as **LDR** Light Dependent Resistor.



Image of LDR or Photo resistor



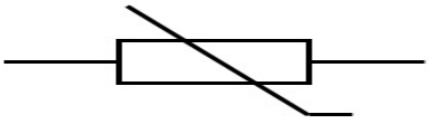
Symbol for LDR or Photo resistor

Varistors: The resistance of a varistor, varies with the applied voltage. As the voltage increases, the resistance decreases and if the voltage decreases, the resistance increases. It is also called

VDR. v



Image of a Varistor



Symbol for Varistor

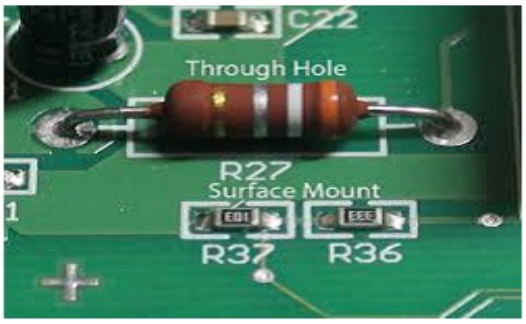
Surface Mount Devices: These are being highly used since the introduction of surface mount technology. These can be termed as **chip resistors**, which means a resistive layer integrated on a ceramic chip.

These surface mount resistors are very small when compared to the normal resistors and hence occupy less space. They are effective and dissipate less heat. The invention of these resistors has changed the look of a PCB PrintedCircuitBoardPrintedCircuitBoard and reduced its size greatly.

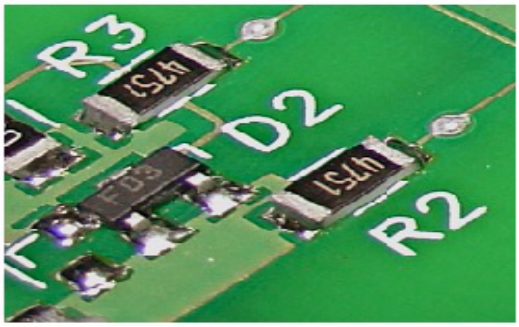
The advantages of surface mount resistors are −

- These are compact in size.
- These are very stable.
- They have good tolerance.
- They are effective in reducing heat dissipation.

The following figure shows the images of surface mount resistors.



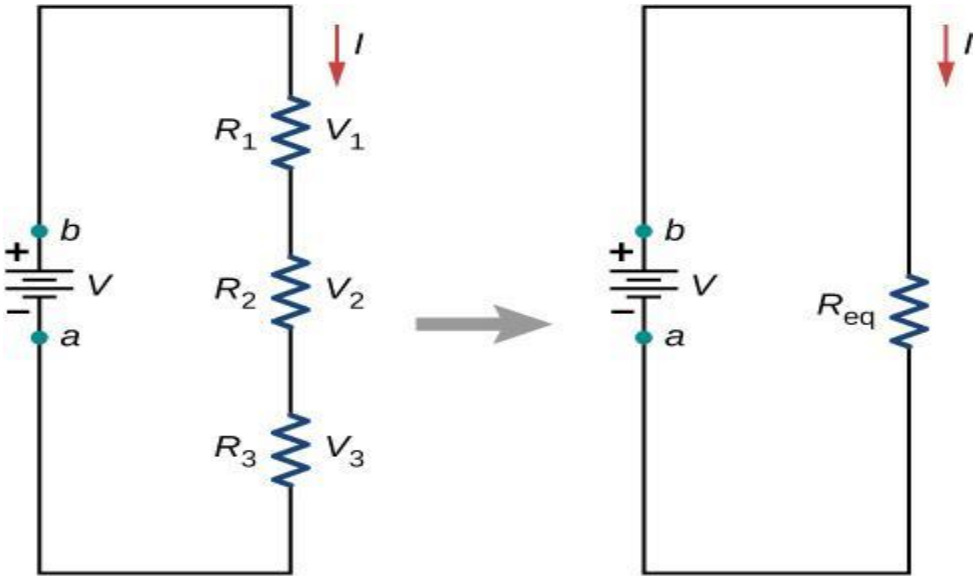
Surface mount resistor when compared to the size of a normal resistor



Surface mount resistors mounted on a PCB

Series and Parallel Combination of resistors:

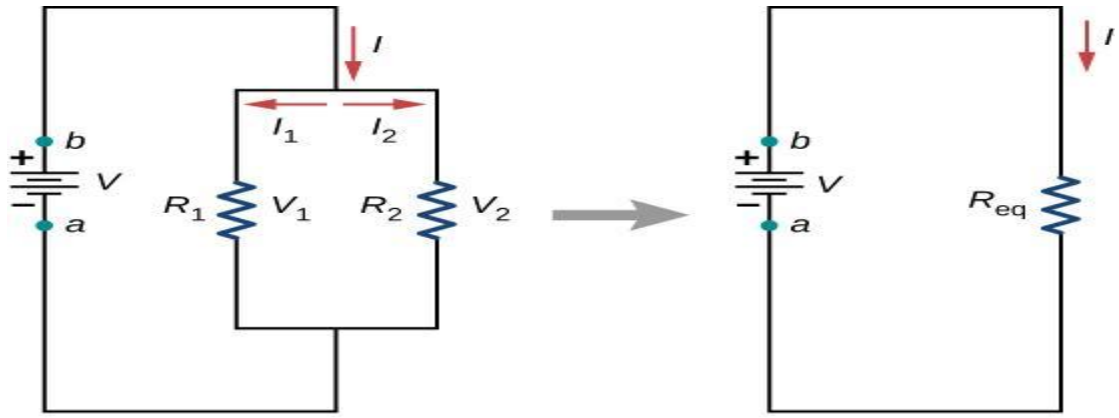
Series Combination:



(a) Original circuit (b) Equivalent circuit

Req= R1+R2+R3+.....

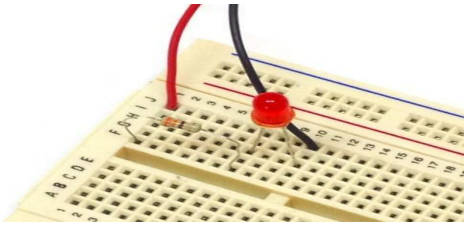
Parallel combination:



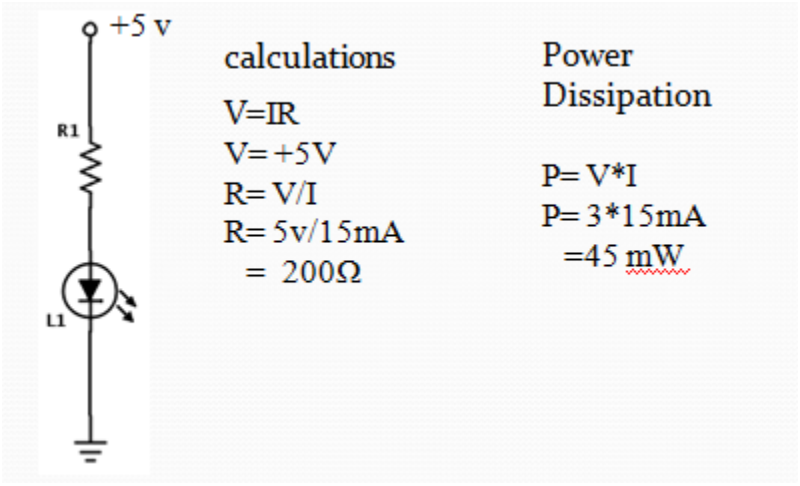
(a) Original circuit (b) Equivalent circuit

$$\frac{1}{R_{Total}} = \frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3} + \ldots + \frac{1}{Rn}$$

Current limiting resistor:

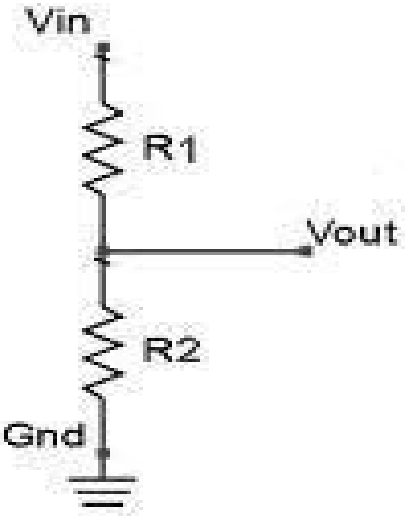


A current limiting resistor is a resistor that is used to reduce the current in a circuit. A simple example is a resistor in series with an LED.



Voltage Division Rule

Circuit:

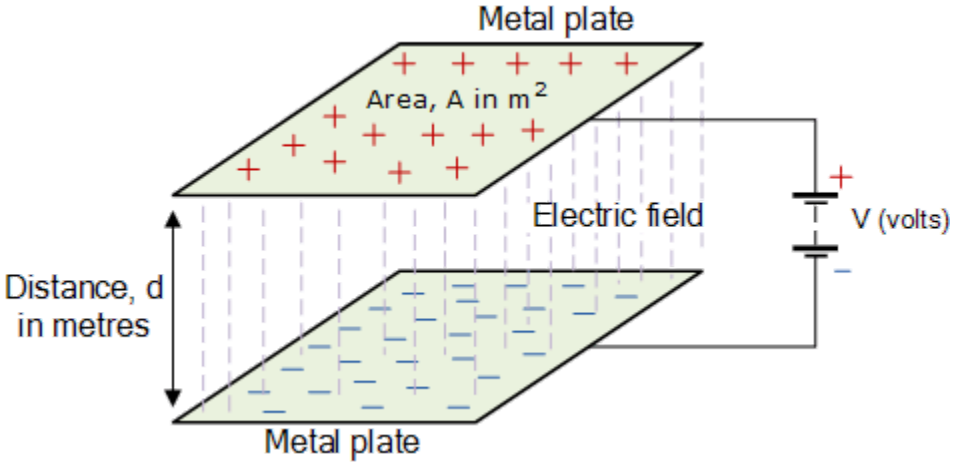


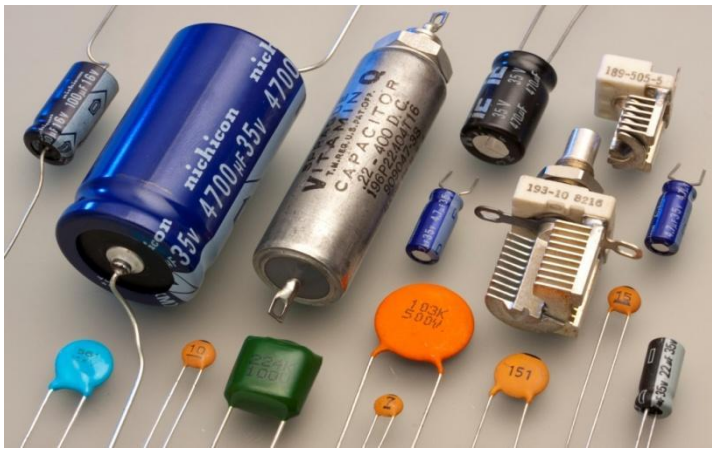
Formula:

$$V_{out} = V_{in} \cdot \frac{R_2}{R_1 + R_2}$$

- **Capacitors:** Passive component that has the ability to store the energy in the form of potential difference between its plates. It

resists a sudden change in voltage. The charge is stored in the form of potential difference between two plates, which form to be positive and negative depending upon the direction of charge storage.A non-conducting region is present between these two plates which is called as dielectric. This dielectric can be vacuum, air, mica, paper, ceramic, aluminum etc. The name of the capacitor is given by the dielectric used.





The capacitance of capacitor: Capacitance is the electrical property of a capacitor and is the measure of a capacitors ability to store an electrical charge onto its two plates with the unit of capacitance being the **Farad** (abbreviated to F) named after the British physicist Michael Faraday. Capacitance is defined as being that a capacitor has the

capacitance of One Farad when a charge of One Coulomb is stored on the plates by a voltage of One **volt**.

- Standard units of capacitor:
- Microfarad (μF) 1μF = 1/1,000,000 = 0.000001 = 10^{-6} F
- Nanofarad (nF) 1nF = 1/1,000,000,000 = 0.000000001 = 10^{-9} F
- Picofarad (pF) 1pF = 1/1,000,000,000,000 = 0.000000000001 = 10^{-12} F

$$C = \frac{\varepsilon A}{d}$$

Where,

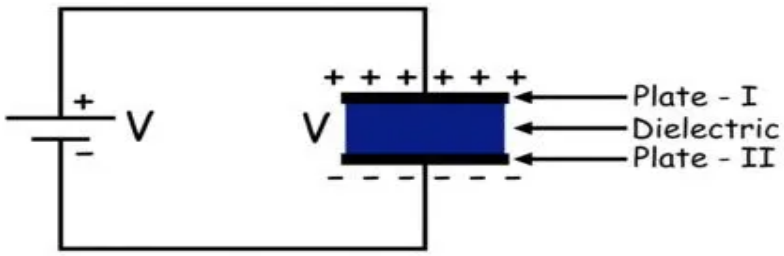
C = Capacitance in Farads

ε = Permittivity of dielectric (absolute, not relative)

A = Area of plate overlap in square meters

d = Distance between plates in meters

Working of Capacitor:



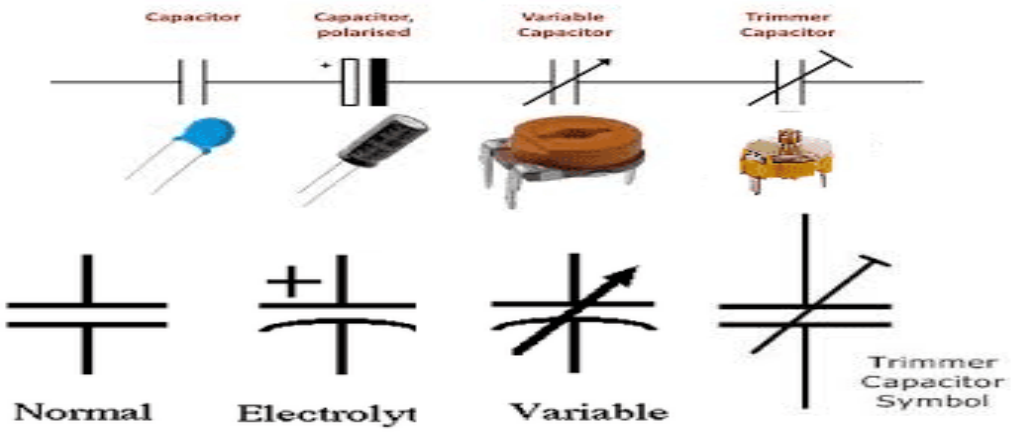
To demonstrate **how does a capacitor work**, let us consider a most basic structure of a capacitor. It is made of two parallel conducting plates

separated by a dielectric that is parallel plate capacitor. When we connect a battery (DC Voltage Source) across the capacitor, one plate (plate-I) gets attached to the positive end, and another plate (plate-II) to the negative end of the battery. Now, the potential of that battery is applied across that capacitor. At that situation, plate-I is in positive potency with respect to the plate-II. At steady state condition, the current from the battery tries to flow through this capacitor from its positive plate (plate-I) to negative plate (plate-II) but cannot flow due to the separation of these plates with an insulating material.

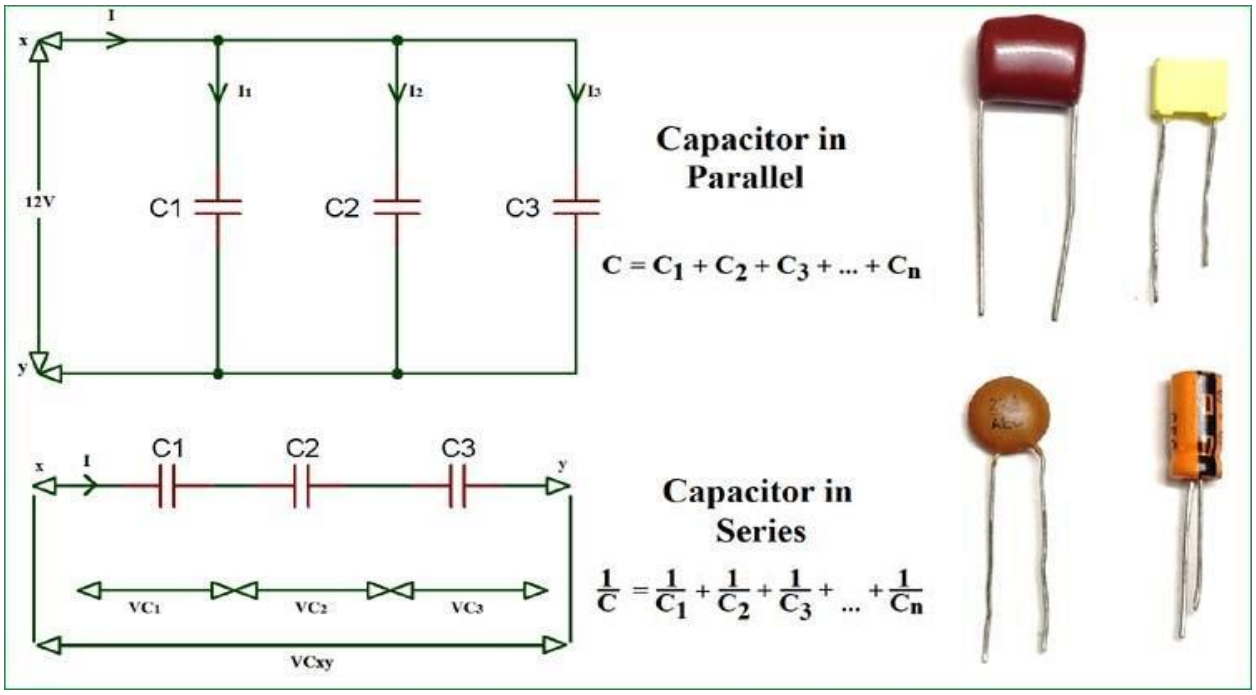
An electric field appears across the capacitor. As time goes on, positive plate (plate I) will accumulate positive charge from the battery, and negative plate (plate II) will accumulate negative charge from the battery. After a certain time, the capacitor holds maximum amount of charge as per its capacitance with respect to this voltage. This time span is called charging time of this capacitor.

After removing this battery from this capacitor, these two plates hold positive and negative charge for a certain time. Thus this capacitor acts as a source electrical energy.

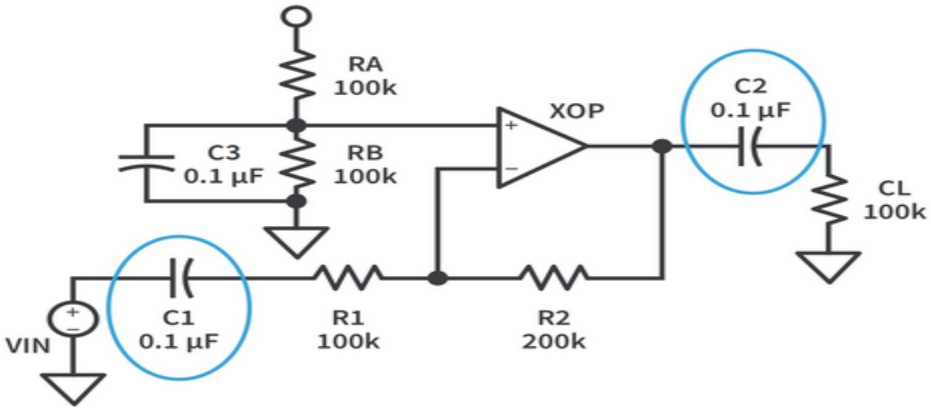
Types of Capacitors:



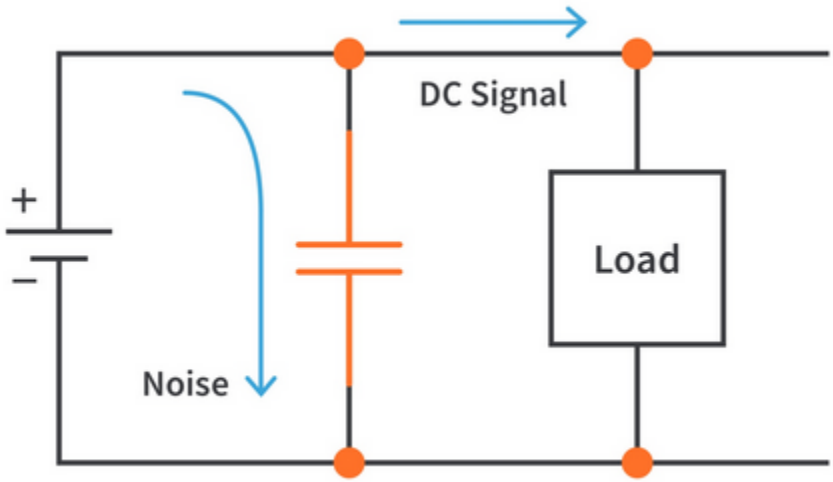
Series and parallel connection of capacitors



Coupling capacitor: Coupling capacitors are connected in series to the signal path. In this way, a coupling capacitor filters DC signals instead of AC signals. Coupling capacitors are widely used in amplifier circuits. Coupling capacitors are usually placed at the input and output of circuit as shown below. They are also placed in between circuit stages.

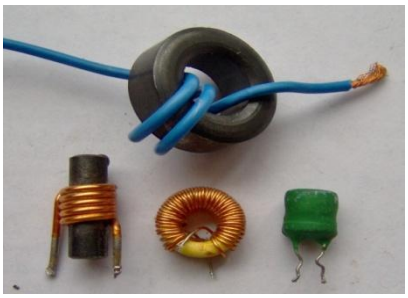


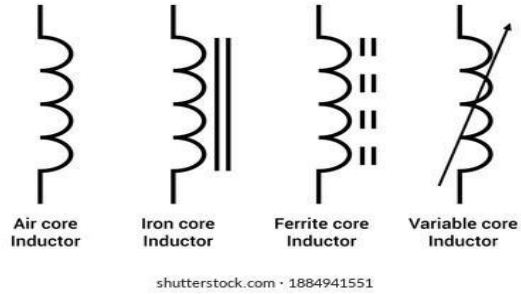
Decoupling Capacitors: A decoupling capacitor is usually used in power supplies and power sections of your circuit. It is placed in parallel with the power source and the load. The first function of a decoupling capacitor is to act as a local electrical energy reservoir. With this, the capacitor can provide energy to keep the voltage stable whenever the input voltage suddenly drops. The second function of a decoupling capacitor is to filter AC noise.



Inductor

An inductor, also called a coil, choke, or reactor. It is passive two-terminal electrical component that stores energy in a magnetic field when electric current flows through it. An inductor typically consists of an insulated wire wound into a coil. The main property of an inductor is that it **opposes any change in current**.





- Unit : The unit of Inductance is **Henry**. It is indicated by **L**. The inductors are mostly available in mH *milliHenry* and µH *microHenry*.

  Inductance Formula:

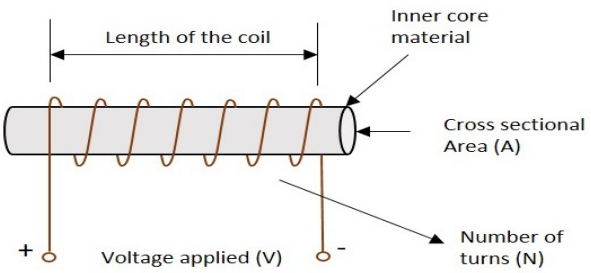
$$L = \frac{\mu N^2 A}{l}$$

Where:

$L = Inductance\ in\ henries\ (H)$
$\mu = permeability\ (Wb/A \cdot m)$
$N = number\ of\ turns\ in\ coil$
$A = area\ encircled\ by\ coil\ (m^2)$
$l = lenth\ of\ coil\ (m)$

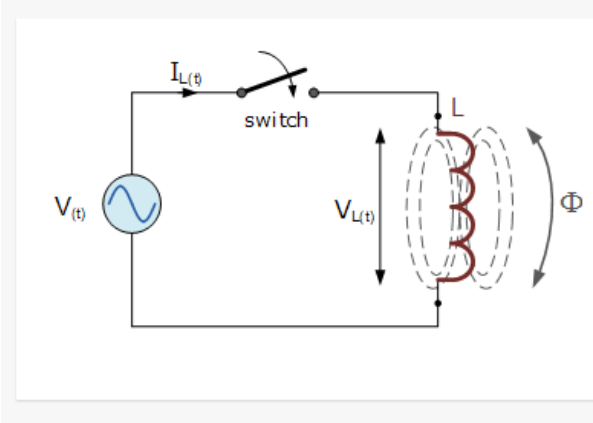
Factors affecting Inductance

- Length of the coil
- Cross sectional area of the coil
- Number of turns
- Permeability of the core

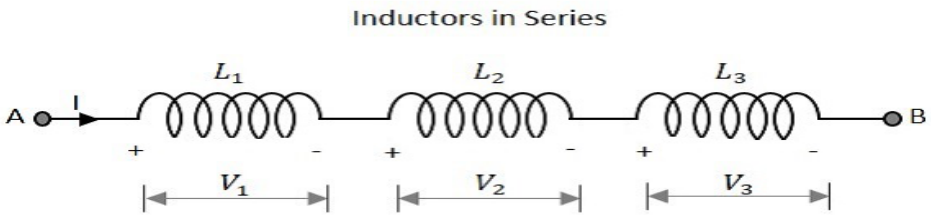


Inductor Working:

Let's start with stating the fact "Magnetic flux will be developed across a current carrying conductor". Likewise when current is applied through an Inductor, it develops magnetic flux around it. In other words energy applied to an Inductor is stored in the form of Magnetic flux. The direction of Magnetic flux developed will be in opposite to the direction of flow of current. Therefore Inductors resist sudden change in current flowing through it. This ability of Inductor is termed as Inductance and every Inductor will have some inductance in it.



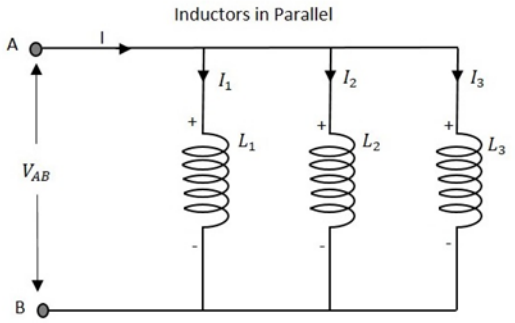
Inductor Connections:



$L=L1+L2+L3$

$V=V1+V2+V3$

I= I1 =I2=I3



$$\frac{1}{l}=\frac{1}{L1}+\frac{1}{L2}+\frac{1}{L3}$$

V= V1=V2=V3

I= I1+I2+I3

Applications of Inductors

There are many applications of Inductors, such as −

Inductors are used in filter circuits to sense high-frequency components and suppress noise signals

To isolate the circuit from unwanted HF signals.

Inductors are used in electrical circuits to form a transformer and isolate the circuits from spikes.

Inductors are also used in motors.

Transformer

Electrical transformer is a static electrical machine which transforms electrical power from one circuit to another circuit, without changing the frequency. Transformer can increase or decrease the voltage with corresponding decrease or increase in current.

The basic principle behind working of a transformer is the phenomenon of mutual induction between two windings linked by common magnetic flux. Basically a transformer consists of two inductive coils; primary winding and secondary winding. The coils are electrically separated but magnetically linked to each other.

When, primary winding is connected to a source of alternating voltage, alternating magnetic flux is produced around the winding. The core provides magnetic path for the flux, to get linked with the secondary winding. Most of the flux gets linked with the secondary winding which is called as 'useful flux' or main 'flux', and the flux which does not get linked with secondary winding is called as 'leakage flux'. As the flux produced is alternating (the direction of it is continuously changing), EMF gets induced in the secondary winding according to Faraday's law of electromagnetic induction. This emf is called 'mutually induced emf', and the frequency of mutually induced emf is same as that of supplied emf. If the secondary winding is closed circuit, then mutually induced current flows through it, and hence the electrical energy is transferred from one circuit (primary) to another circuit (secondary).

Types of transformer:

On the basis of their purpose

1. Step up transformer: Voltage increases (with subsequent decrease in current) at secondary.
2. Step down transformer: Voltage decreases (with subsequent increase in current) at secondary.

On the basis of type of supply

1. Single phase transformer

2. Three phase transformer

On the basis of their use

1. Power transformer: Used in transmission network, high rating
2. Distribution transformer: Used in distribution network, comparatively lower rating than that of power transformers.
3. Instrument transformer: Used in relay and protection purpose in different instruments in industries
    - Current transformer (CT)
    - Potential transformer (PT)

Turns Ratio

- The ratio of number of turns in the primary coil to the number of turns in the secondary coil is called as the "turns ratio" or "the ratio of transformation". The turns ratio is usually denoted by N.
- ***N=Turns ratio=Number of turns on Primary /Number of turns on Secondary***

=Np/Ns

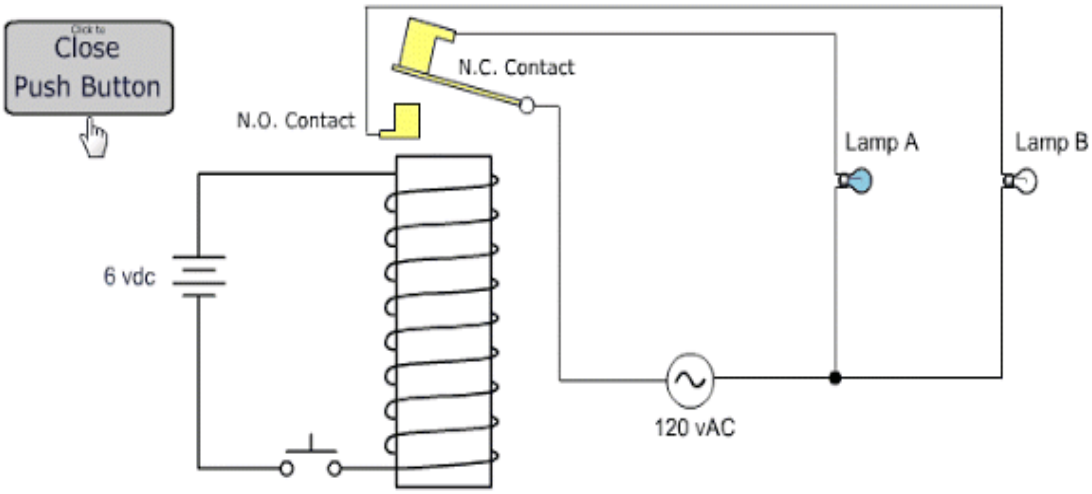
- ***Np/Ns=Vp/Vs=N=Turns ratio***

Relay:

A Relay is an electromechanical device that can be used to make or break an electrical connection**.** It consists of a flexible moving mechanical part which can be controlled electronically through an electromagnet.

The traditional form of a relay uses an electromagnet to close or open the contacts, but other operating principles have been invented, such as in solid-state relays which use semiconductor

properties for control without relying on moving parts. Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from overload or faults; in modern electric power systems these functions are performed by digital instruments still called *protective relays*.



Relay works on the principle of electromagnetic induction. When the electromagnet is applied with some current it induces a magnetic field around it. Above image shows working of the relay .A switch is used to apply DC current to the load. In the relay Copper coil and the iron core acts as electromagnet. When the coil is applied with DC current it starts attracting the contact as shown. This is called energizing of relay. When the supply is removed it retrieves back to the original position. This is called De energizing of relay. There are also such relays, whose contacts are initially closed and opened when there is supply i.e. exactly to opposite to the above shown relay.

Solid state relays will have sensing element to sense the input voltage and switches the output using opto-coupling.

Relay Contact Types: As we have seen that relay is a switch. The terminology “Poles and throws” is also applicable for relay. Depending

on the number of contacts and number of circuits it switches relays can be classified.

Poles and Throws

Depending on the poles and throws, relays are classified into

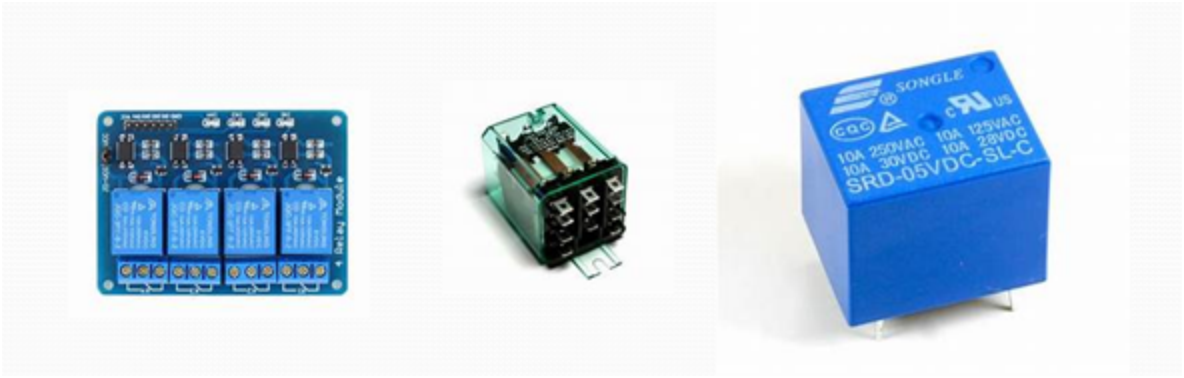
- Single pole single throw
- Single pole double throw
- Double pole single throw
- Double pole double throw

Relay Applications

Relays are used to protect the electrical system and to minimize the damage to the equipment connected in the system due to over currents/voltages. The relay is used for the purpose of protection of the equipment connected with it.

These are used to control the high voltage circuit with low voltage signal in applications audio amplifiers and some types of modems. Typical application areas of the relays include

- Lighting control systems
- Telecommunication
- Industrial process controllers
- Traffic control
- Motor drives control
- Protection systems of electrical power system
- Computer interfaces
- Automotive
- Home appliances



Active Components

An active component is an electronic component which supplies energy to a circuit. Common examples includesVoltage sources, Current sources, Generators (such as alternators and DC generators),all different types of transistors (such as bipolar junction transistors, MOSFETS, FETs, and JFET),Diodes (such as Zener diodes, photodiodes, Schottky diodes, and LEDs)

Voltage Source:

Ideal and practical voltage source

Voltage source that produces constant voltage across its terminals, no matter what amount of current is drawn from it. Terminal voltage is independent of load resistance.

Practical voltage source :In this as load resistance is decreased ,corresponding load current increases while terminal voltage across decreases.

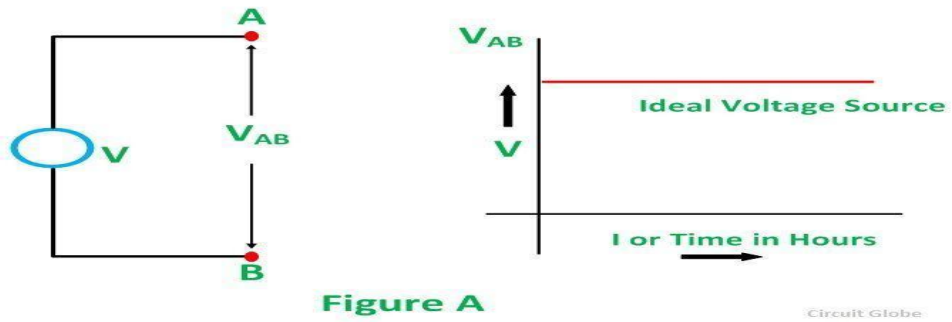


Figure A

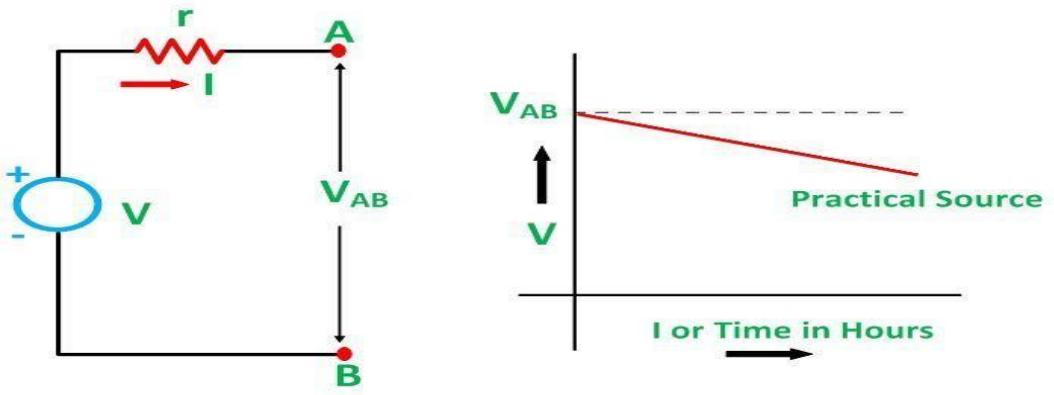


Figure B

Current Source:

An **ideal current source** is a two-terminal circuit element which supplies the same current to any load resistance connected across its terminals. It is important to keep in mind that the current supplied by the current source is independent of the voltage of source terminals. It has infinite resistance.

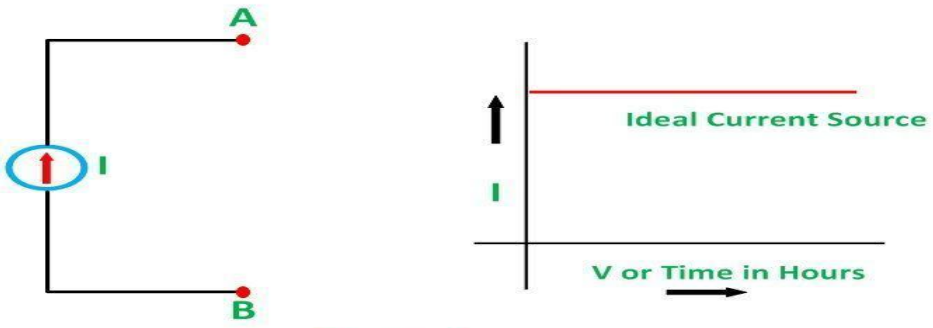


Figure C

Practical Current Source

A practical current source is represented as an ideal current source connected with the resistance in parallel. The symbolic representation is shown below:

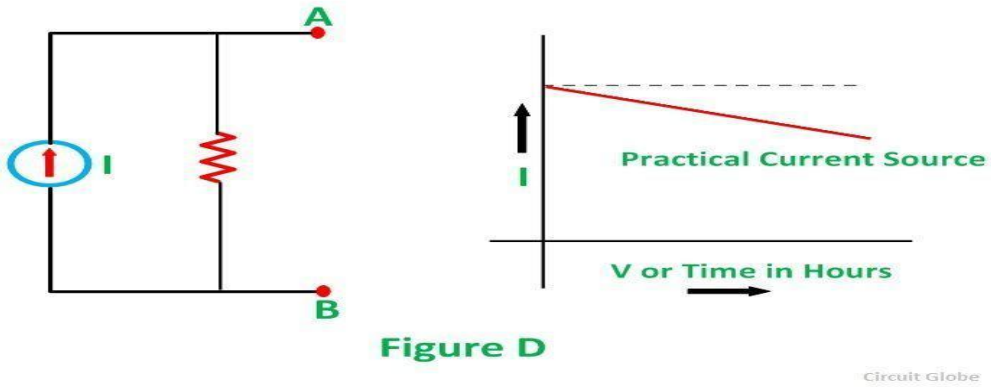
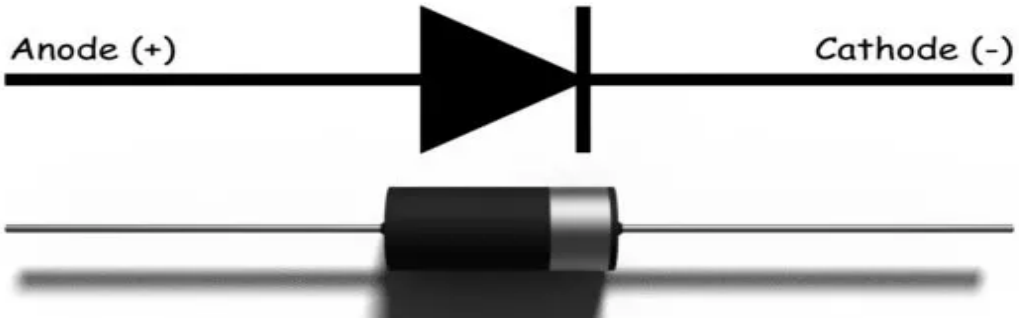


Figure D

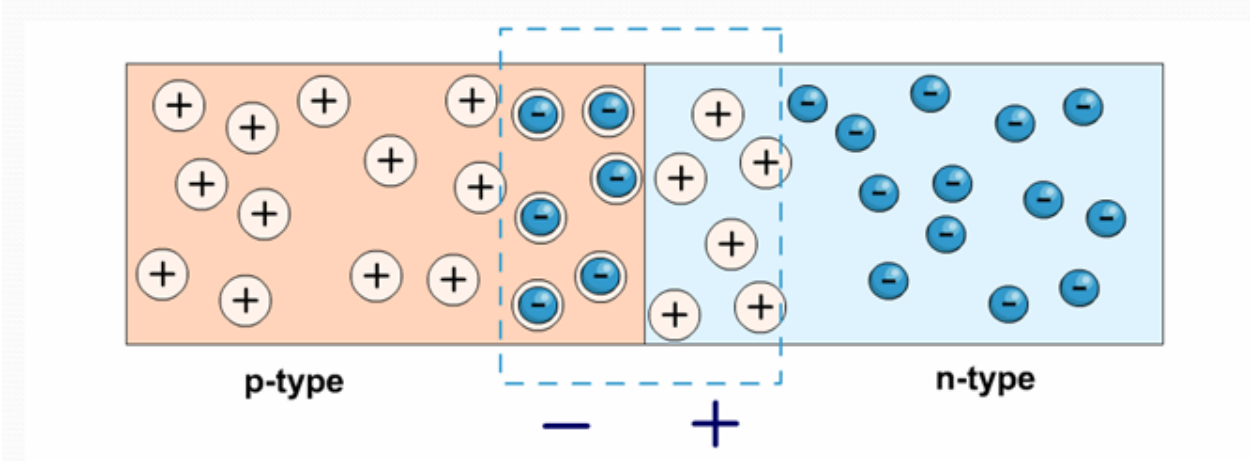
Diode

A **diode** is a semiconductor device that essentially acts as a one-way switch for current. It allows current to flow easily in one direction, but severely restricts current from flowing in the opposite direction.

Symbol:



working :



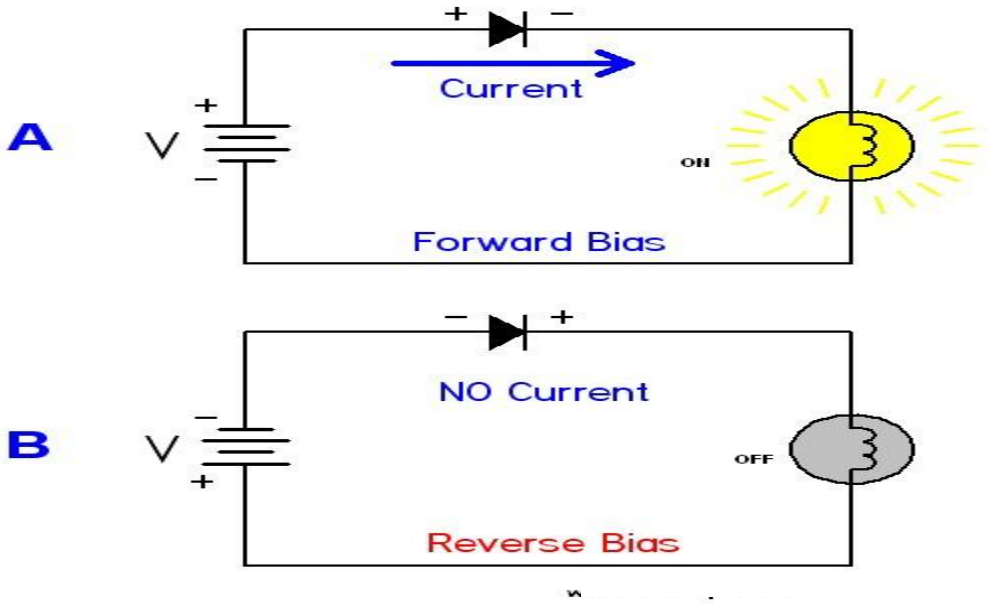
The most common type of diode uses a *p-n* junction. In this type of diode, one material (*n*) in which electrons are charge carriers abuts a second material (*p*) in which holes (places depleted of electrons that act as positively charged particles) act as charge carriers. At their interface, a depletion region is formed across which electrons diffuse to fill holes in the *p*-side. This stops the further flow of electrons.

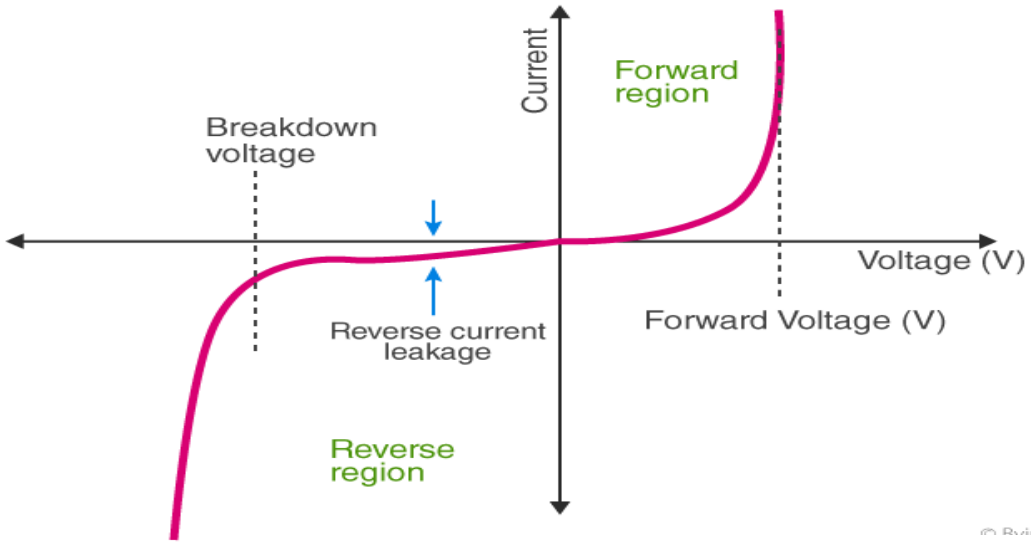
When this junction is forward biased (that is, a positive voltage is applied to the *p*-side), electrons can easily move across the junction to fill the holes, and a current flows through the diode.

When the junction is reverse biased (that is, a negative voltage is applied to the *p*-side), the depletion region widens and electrons cannot easily move across. The current remains very

small until a certain voltage (the breakdown voltage) is reached and the current suddenly increases.

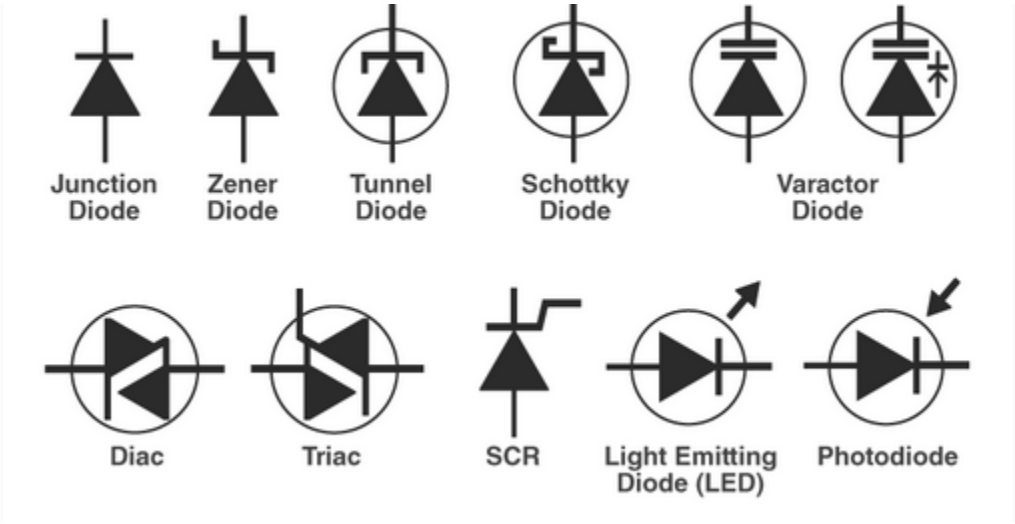
Forward and reverse bias connections and their V-I characteristics are as shown below.





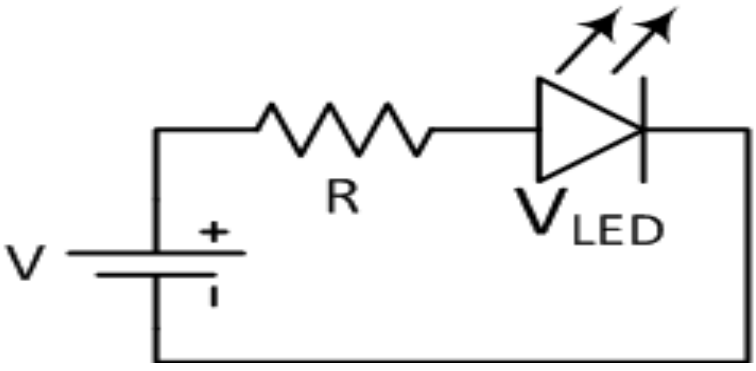
Types of diodes

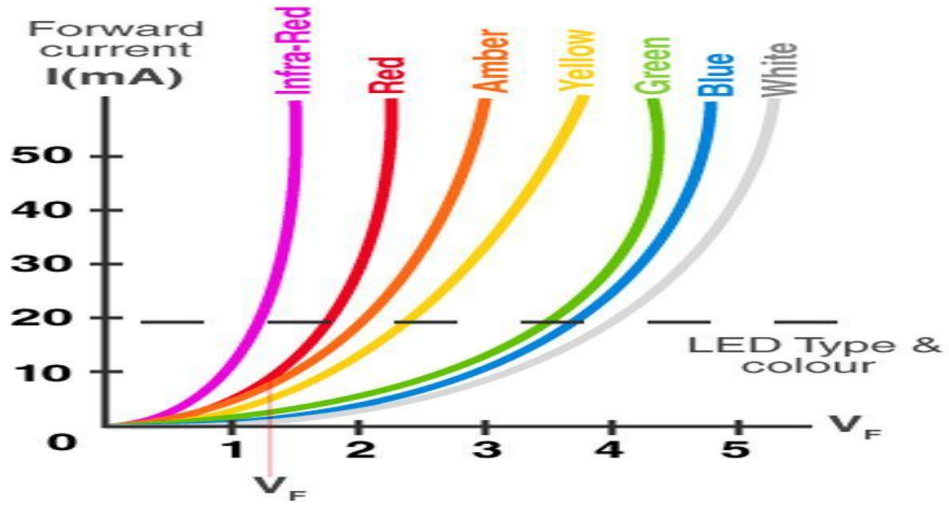
1. Light Emitting Diode
2. Laser diode
3. Avalanche diode
4. Zener diode
5. Schottky diode
6. Photodiode
7. PN junction diode



Light Emitting Diode:

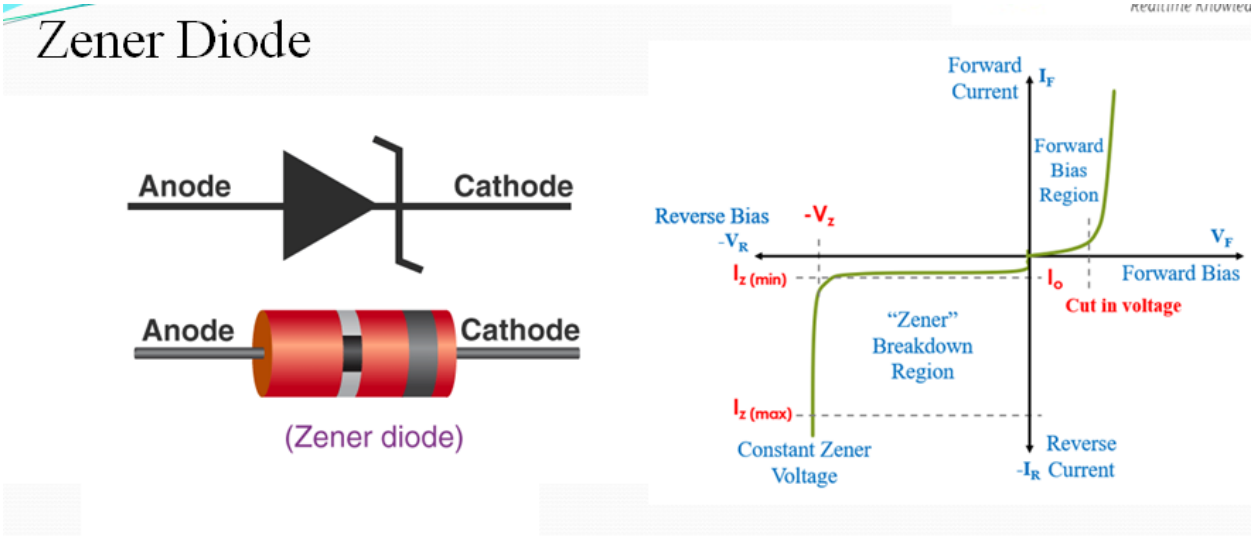
An LED is basically a PN Junction Diode, which emits light when forward biased. LEd's are available in different colours. Light colour depends on the energy gap of the semiconductor used.



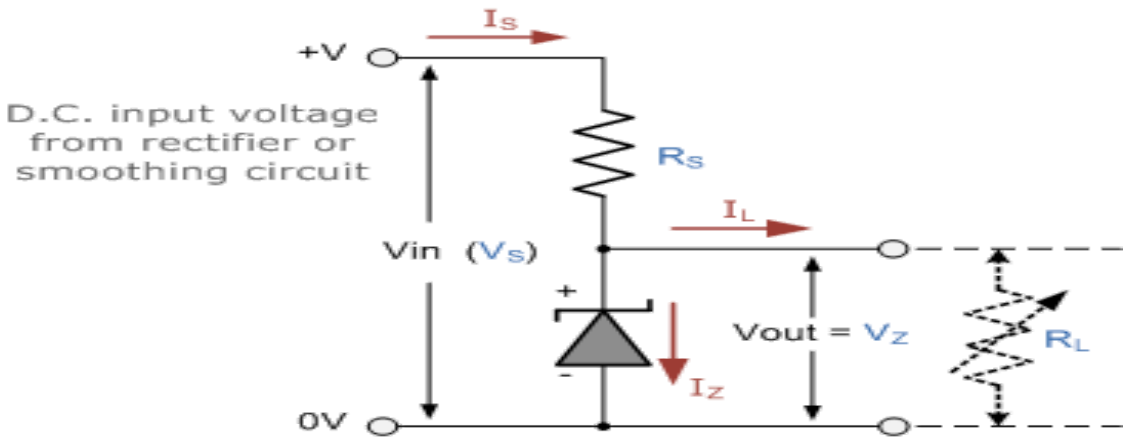


Zener Diode

It is the most useful type of diode as it can provide a stable reference voltage. These are operated in reverse bias and break down on the arrival of a certain voltage. If current passing through the resistor is limited, a stable voltage is generated. Zener diodes are widely used in power supplies to provide a reference voltage. By adjusting doping level ,zener diodes of different voltages from 2 to 200 V and of different wattages is manufactured.They are categorized by power dissipation, nominal working voltage, forward current, forward voltage, packaging type and maximum reverse current.



Zener as Voltage Regulator



working:

A voltage regulator is an electronic circuit that provides a stable DC voltage independent of the load current, temperature and AC line voltage variations. A Zener diode of break down voltage *VZ* is reverse connected to an input voltage source *VI* across a load resistance *RL* and a series resistor *RS*. The voltage across the zener will remain steady at its break down voltage *VZ* for all the values of zener current *IZ* as long as the current remains in the break down region. Hence a regulated DC output voltage *V*0=*VZ* is obtained across *RL*, whenever the input voltage remains within a minimum and maximum voltage.

Diode Applications

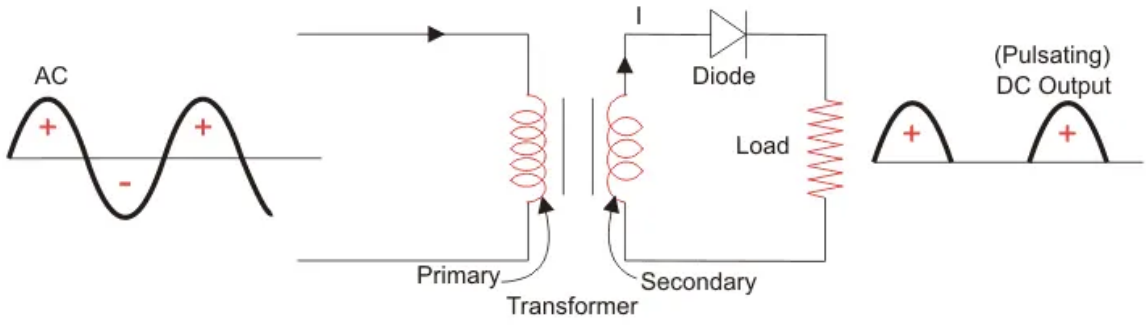
- Rectifying a voltage
- Isolating signals from a supply
- Logic gates
- Over voltage protections
- Voltage multiplier circuits

Rectifier Circuits

A rectifier is a device that converts alternating current (AC) to direct current (DC). It is done by using a diode or a group of diodes. Half wave rectifiers use one diode, while a full wave rectifier uses multiple diodes.

Half Wave rectifier:

A half wave rectifier is the simplest form of rectifier available. The diagram below illustrates the basic principle of a half-wave rectifier. When a standard AC waveform is passed through a half-wave rectifier, only half of the AC waveform remains. Half-wave rectifiers only allow one half-cycle of the AC voltage through and will block the other half-cycle on the DC side, as seen below.

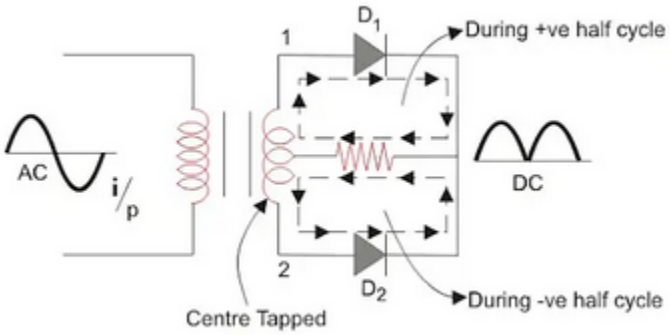


During the positive half cycle of the AC voltage, the diode will be forward biased and the current flows through the diode.

During the negative half cycle of the AC voltage, the diode will be reverse biased and the flow of current will be blocked. The final output voltage waveform on the secondary side (DC) is shown in figure above.

Full Wave Rectifier

The full wave rectifier circuit has two diodes and a step down transformer that are linked and are centre tapped. Across the load resistor we get the output voltage.



AC voltage is supplied to the input transformer. When there is the positive half cycle of the AC voltage then terminal 1 will be positive, terminal two will be negative and centre tap will be at the zero potential.

At the time of the positive half cycle the diode D1 will be forward biased and diode D2 will be reverse biased. D1 will let the current flow and D2 will block the flow through it.

In case of a negative half cycle, the diode D1 is reversed biased and the diode D2 is forward biased. This is because the top half of the secondary circuit becomes negative, while the bottom half of it becomes positive. Therefore, in a full wave rectifier, DC voltage is obtained for both positive as well as negative half cycle.

Bridge Rectifier

Bridge Rectifiers use four diodes that are arranged cleverly to convert the AC supply voltage to a DC supply voltage. The output signal of such a circuit is always of the same polarity regardless of the polarities of the input AC signal. Figure depicts the circuit of a bridge rectifier with diodes interlocked in a bridge configuration. The AC signal is applied at the input terminals a and b, and the output is observed across the load resistor R1.

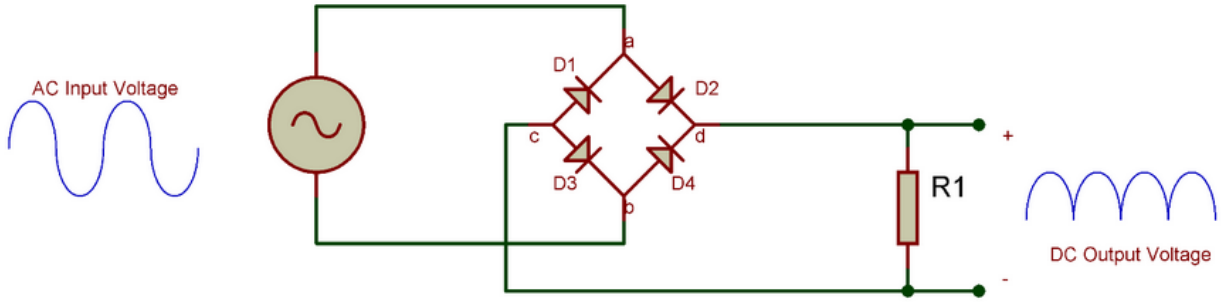


Figure 2 Bridge Rectifier with load resistor

working:

1. In the first positive half cycle of the AC signal, the diodes D2 and D3 become forward biased and start conducting. At the same time, the diodes D1 and D4 will be reverse biased and will not conduct. The current will flow through the load resistor via the two forward-biased diodes. The voltage seen at the output will be positive at terminal d and negative at terminal c.
2. Now, during the negative half cycle of the AC signal, the diodes D1 and D4 will be forward biased and diodes D2 and D3 will become reverse biased. The positive voltage will appear on the anode of D4, and negative voltage will be applied to the cathode of D1. It is worth noting at this point that the current that will be flowing through the load resistor will have the same direction as it has with the positive half cycle. Therefore, no matter the polarity of the input signal, the output polarity will always be the same. We

can also say that the negative half cycle of the AC signal has been inverted and is appearing as a positive voltage at the output.

Applications of Bridge rectifier

- For supplying polarized and steady DC voltage in welding.
- Inside power supplies
- Inside battery chargers
- Inside wind turbines
- To detect the amplitude of modulating signals
- For conversion of high AC to low DC voltage